

STA 209 Fish Quiz

1. What are we plotting?

You repeatedly take samples of 10 fish and record each sample's mean length. After 500 samples, you make a histogram of those means.

- What does one observation in this histogram represent?
- How many observations are in this histogram?

2. Predict the change.

You increase the number of fish **in each sample** from 10 to 40, keeping the population unchanged. What happens to:

- The center of the sampling distribution?
- The standard error: does it stay the same, halve, or become one-quarter as large?

3. More samples or larger samples?

Two students each simulate samples of size ($n=10$) from the same population. One generates 100 sample means; the other generates 10,000 sample means.

Does the second student's sample mean have a smaller theoretical standard error? Explain briefly what generating more sample means accomplishes.

4. Give me one fish fact

It can be whatever. It doesn't even have to be true.